复流形上的几何 III

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January 7, 2026

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1 Lecture 8

Submersion of complex manifolds

The background of this section is actually the context of family local index theorem of Bismut. The necessary geometric data are listed as below.

1. Let $\pi : M \to B$ be a holomorphic submersion of complex manifolds, with compact fiber Z. Then for any $b \in B$, there is a neighborhood U of b such that $\pi^{-1}(U)$ is diffeomorphic to $U \times Z$.

2. Let $T^H M$ be a horizontal subbundle of TM, i.e., $T_x M = T_x^H M \oplus T_x Z$. Clearly, $\pi_* : T_x^H M \to T_b B$ is an isomorphism for any $x \in \pi^{-1}(b)$, and $T_x^H M \cong \pi^* T_b B$.

3. Let g^{TZ} be a metric on the holomorphic vector bundle TZ. Since $T_x Z$ is the kernel of $\pi_* : T_x M \to T_b B$, it is a complex subspace of $T_x M$. Thus the complex structure J on TM preserves TZ. Let $J^{TZ} := J|_{TZ}$ be the restriction of J to TZ. We assume that g^{TZ} is a J^{TZ}-invariant metric, i.e.,

$$g^{TZ}(J^{TZ}u, J^{TZ}v) = g^{TZ}(u, v), \quad \forall u, v \in TZ.$$

We can extend g^{TZ} $\mathbb{C}$-bilinearly, still denoted by g^{TZ}. We last assume that J preserves the horizontal subbundle $T^H M$, i.e., $J(T^H M) = T^H M$.

Definition 1.1. *We say a triple $(\pi, g^{TZ}, T^H M)$ is a Kahler fibration, if there exists real, closed $\omega \in \Omega^{1,1}(M)$ such that*

- *$T^H M \perp_\omega TZ$, i.e., $\omega(u, y) = 0$ for any $u \in T^H M$, $y \in TZ$;*
- *$\omega|_{TZ}$ is a Kä hler form on each fiber $Z_b = \pi^{-1}(b)$, i.e.,*

$$\omega(X, Y) = g^{TZ}(X, JY), \quad \forall X, Y \in TZ.$$